# 22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

## Unit-IV: MODERN PHYSICS_LM6

### Dual Nature of Light and the De Broglie Hypothesis

**1. Dual Nature of Light:**

The dual nature of light refers to its ability to exhibit both wave-like and particle-like properties. This concept was first proposed by Thomas Young in his famous double-slit experiment, where light exhibited interference patterns characteristic of waves.

Wave Properties: Light exhibits phenomena such as interference, diffraction, and polarization, which are characteristic of wave behavior.

Particle Properties: The photoelectric effect, where light ejects electrons from a material surface, and the Compton effect, where light scatters off particles, demonstrate particle-like behavior of light.

2. De Broglie Hypothesis:

Louis de Broglie proposed that if light can exhibit both wave and particle properties, then particles such as electrons and atoms should also exhibit wave-like behavior. This hypothesis introduced the concept of matter waves.

De Broglie Wavelength: De Broglie suggested that every particle has an associated wavelength, known as the de Broglie wavelength.

## Dual Nature of Matter (Wave-Particle Duality)

**According to Louis de Broglie (in 1923), the matter has dual nature**

**All the matter particles like electron and neutron have an associated wave with them (Matter waves or de Broglie waves)**

**Matter wave can behave both as a wave-like and as a particle-like characteristics(but they will exhibit only one property at atime)**

- **Wave-Particle duality holds that light and matter exhibit properties of both waves and particles**
- **Photoelectric Effect proves the particle nature of light**
- **Davisson-Germer experiment showed the wave nature of matter (diffraction of electrons)**

### Expression for de-Broglie Wavelength

Energy of a particle (Einstein's theory)

$$E = mc^2$$

$m$ – mass of the particle
$c$ – velocity of light

Energy of a photon (Planck's theory)

$$E = h\nu$$

$h$ – Planck's constant
$\nu$ – Frequecy of radation

$$mc^2 = h\nu$$

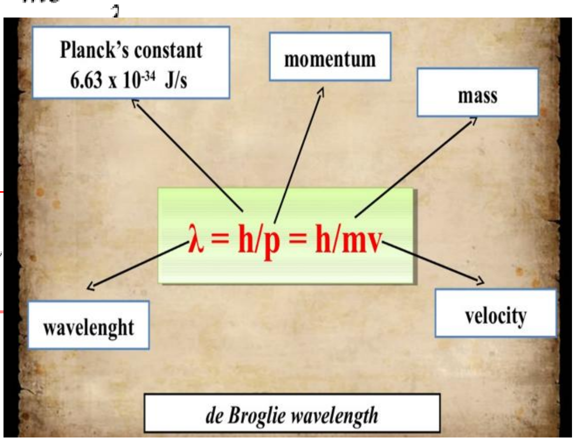
$$mc \cdot c = h\nu$$

$$mc = \frac{h\nu}{c}$$

$$\left[c = \nu\lambda \rightarrow \frac{\nu}{c} = \frac{1}{\lambda}\right]$$

$$mc = \frac{h}{\lambda}$$

$\lambda$




**de Broglie Wavelength**

The de Broglie equation relates a moving particle's wavelength with its momentum

All matter exhibits properties of both particles and waves

This is true only if the particle is very light like an electron

## de-Broglie Wavelength in terms of Energy (E) & Voltage (V)

Kinetic Energy of a moving particle is

$$E = \frac{1}{2}mv^2$$

$$E = \frac{1}{2m}m^2v^2$$

$$E = \frac{p^2}{2m}$$

$$p^2 = 2mE \rightarrow p = \sqrt{2mE}$$

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}}$$

An electron of mass 'm' and charge 'e' is accelerated by a potential 'V'

Energy acquired by the electron 'eV' is equal to the kinetic energy ' ½ mv² '

$$E = \frac{1}{2}mv^2 = eV$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\boxed{\lambda = \frac{h}{\sqrt{2meV}}}$$

**Wave Nature of Particles and Basics of Davisson-Germer Experiment**

**Overview:**

The Davisson-Germer experiment provided direct experimental evidence for the wave-like nature of electrons, confirming de Broglie's hypothesis. In this experiment, electrons were diffracted by a crystal lattice, producing interference patterns characteristic of waves.

Experimental Setup: In the Davisson-Germer experiment, a beam of electrons is directed at a crystalline target. The electrons are diffracted by the regular arrangement of atoms in the crystal lattice, resulting in an interference pattern on a detector screen.

Interference Pattern: The observed interference pattern arises from the constructive and destructive interference of the electron waves as they pass through the crystal lattice. The pattern confirms the wave-like nature of electrons and validates de Broglie's hypothesis.

**Key Results:** The Davisson-Germer experiment demonstrated that electrons exhibit diffraction patterns similar to those of light waves, providing compelling evidence for the wave-particle duality of matter.

**Example: Electron Diffraction in Crystals**

Consider a beam of electrons incident on a crystalline material. The regular arrangement of atoms in the crystal lattice acts as a diffraction grating, causing the electrons to diffract. The resulting diffraction pattern on a detector screen exhibits interference fringes, indicating the wave-like behavior of electrons.

**Example: Electron Microscopy**

Transmission electron microscopy (TEM) relies on the wave nature of electrons to visualize nanoscale structures. In TEM, a beam of electrons is transmitted through a thin specimen, and the resulting diffraction pattern is used to reconstruct the specimen's internal structure with high resolution.

Conclusion:

The dual nature of light, as well as the wave-particle duality of matter, are fundamental principles of quantum mechanics. The De Broglie hypothesis and the Davisson-Germer experiment provide compelling evidence for the wave-like behavior of particles, further challenging classical notions of particle physics. Understanding these concepts is essential for grasping the fundamental nature of particles and the profound implications of quantum mechanics.